THE SUPPORT OF AN EFFECTIVE RELATIVE CYCLE IS CLOSED

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Effectivity is preserved by base change, and the support of a family of cycles behaves well under base change by families of morphisms (relative-cycles chapter, Lemma lemma-effective-functoriality, canonical tag 0H5S, and Lemma lemma-support-family, canonical tag 0H4V). cited at proof lines 1947-1950. Reduces the statement to irreducible components of the base viewed as integral closed subschemes.
- Specializations in a locally Noetherian scheme can be realized by a morphism from the spectrum of a discrete valuation ring (Properties chapter, Lemma properties-lemma-locally-Noetherian-specialization-dvr, canonical tag 054F). cited at proof lines 1953-1957. Realizes the specialization from the generic point eta to any point s, reducing to the valuation ring case.
- Dimension of fibres over a valuation ring dominated by an integral scheme (Varieties, Lemma varieties-lemma-dominate-valuation-ring-dimension-fibres, canonical tag 0B2J). cited at proof lines 1970-1972. The closed fibres of the closures Z-bar_i are equidimensional of dimension r, so the positive coefficients detect the whole support.

Source reference. Original lemma 0H5S: sources/stacks-chow/tags.txt, source lines 21397–21397. Cited at proof lines 1947-1948: reduces to the base change by the inclusion of an irreducible component.

Source reference. Original lemma 0H4V: sources/stacks-chow/tags.txt, source lines 21365–21365. Cited at proof lines 1947-1948: supports pull back along the component inclusion.

Source reference. Original lemma 054F: sources/stacks-chow/tags.txt, source lines 6638–6638. Cited at proof lines 1953-1957: realizes the specialization from the generic point to any point.

Source reference. Original lemma 0B2J: sources/stacks-chow/tags.txt, source lines 13932–13932. Cited at proof lines 1970-1972: the closed fibres Z-bar_{i,0} are equidimensional of dimension r.

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Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Assume S locally Noetherian and f locally of finite type. Let $r, e \geq 0$ be integers. Let α be a relative r-cycle on X/S. If α is effective, then Supp(α) is closed in X.*

Proof. Let $g : S' \to S$ be the inclusion of an irreducible component viewed as an integral closed subscheme. By Lemmas 0H5S and 0H4V it suffices to show that the support of the base change $g^*\alpha$ is closed in $S' \times_S S$. Thus we may assume S is an integral scheme with generic point η. We will show that $\mathrm{Supp}(\alpha)$ is the closure of $\mathrm{Supp}(\alpha_\eta)$. To do this, pick any $s \in S$. We can find a morphism $g : S' \to S$ where S' is the spectrum of a discrete valuation ring mapping the generic point $\eta' \in S'$ to η and the closed point $0 \in S'$ to s, see Properties, Lemma 054F. Then it suffices to prove that the support of $g^*\alpha$ is equal to the closure of $\mathrm{Supp}((g^\alpha)_{\eta'})$. This reduces us to the case discussed in the next paragraph.

Here S is the spectrum of a discrete valuation ring with generic point η and closed point 0. We have to show that $\mathrm{Supp}(\alpha)$ is the closure of $\mathrm{Supp}(\alpha_\eta)$. Since α is effective we may write $\alpha_\eta = \sum n_i[Z_i]$ with $n_i > 0$ and $Z_i \subset X_\eta$ integral closed of dimension r. Since $\alpha_0 = sp_{X/S}(\alpha_\eta)$ we know that $\alpha_0 = \sum n_i[\overline{Z}_{i,0}]_r$ where $\overline{Z}_i$ is the closure of Z_i. By Varieties, Lemma 0B2J we see that $\overline{Z}_{i,0}$ is equidimensional of dimension r. Since $n_i > 0$ we conclude that $\mathrm{Supp}(\alpha_0)$ is equal to the union of the $\overline{Z}_{i,0}$ which is the fibre over 0 of $\bigcup \overline{Z}_i$ which in turn is the closure of $\bigcup Z_i$ as desired. $\square$